TAKE-HOME ASSIGNMENT ·

Growify Digital - Data Analyst + AI Developer Test

You are joining a marketing agency. This test mirrors our real data stack — raw campaign and Sales data flows through Python and into SQL, which becomes the single source of truth powering both a Power BI dashboard and an AI insight tool.

Python · SQL · Power BI	**Power BI Desktop required**	**SQLite or PostgreSQL**	**GitHub repo or ZIP**

DOMAIN KNOWLEDGE & EXPECTED OUTPUT — READ THIS BEFORE YOU START

We are Growify, a performance marketing agency that helps brands scale through data-driven decision-making and performance-focused campaigns. Before building anything, it's important to understand how we approach campaign data. Your SQL schema, Power BI dashboard, and AI outputs should align with this mental model.

Domain Knowledge & Expected Output guide Link - [Click here]

THE PIPELINE YOU MUST BUILD

SQL is the backbone of this architecture. Power BI and the AI tool must both read from the same database — not from separate CSV files. This is what we evaluate most closely: does your pipeline hold together end to end?

Raw CSV
raw.csv · messy ad platform export

↓

Task 1 — Python
clean · validate · recalculate metrics

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Task 2 — SQL database
single source of truth · both outputs query here

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Task 3 — Power BI	**Task 4 — AI insight tool**
SQL query → dashboard + DAX	SQL query → LLM context → answer
↓	↓
Visual reports for clients	Natural language insights

★ SQL is the backbone. Power BI and the AI tool must both read from the same database — not from CSV files.

THE DATASET

You will receive campaigns_raw.csv and Shopify_Raw.csv mimicking a real ad and sales platform export, with intentional quality issues baked in.

Dataset Link - [Raw_Data.csv]

⚠ Intentional issues: duplicate rows, inconsistent date formats, missing spend/click values, wrong CTR values (doesn't match clicks ÷ impressions), mixed-case platform names, outlier spend values. Find and fix them — we don't say where.

TASKS

TASK 1

Purpose: We are a High Tech Enabled agency and we want to know your grasping skills for using tool.

- Step 1 - Please click on the following link https://airtable.com/invite/r/4Jy8m50k
- Step 2 - Sign up and do the onboarding (don't worry about choosing the right setups here)
- Step 3 - Verify the email by going to your email and confirm the email
- Step 4 - Setup a base Add 3 columns with 3 rows of data
    - Name
    - Number
    - Lead Priority (Single Select)
    - High Value
    - Med Value
    - Low Value
- Step 5 - Create a new view grid video and in this group the data by Lead Priority
- Step 6 - Take a screenshot of this view and share it with us

TASK 2

Python — data cleaning & validation

Python / pandas

Clean the raw CSV and load it into a SQL database. This is the foundation everything else depends on — quality here determines quality everywhere downstream.

→ Identify and remove duplicate rows — explain how you detected them
→ Standardise all date formats and validate start/end date lo gic
→ Handle missing values with a clear, justified strategy (don't just drop everything)
→ Recalculate CTR, CPM, ROI and CPC from source columns; flag rows where originals were wrong
→ Normalise all string columns — platform, channel, region, status
→ Load the cleaned data into a SQL database (SQLite is fine; PostgreSQL preferred)
→ Write a brief data quality report documenting every issue found and fixed

→ Output: cleaned_campaigns.db and cleaned_Shopify.db (or equivalent) and data_quality_report.md

TASK 3

SQL — schema design & query layer

SQL

Design a clean SQL schema and write the queries that will feed Power BI and the AI tool. Both downstream tools must pull from here, not from any CSV.

- → Design a clean table schema with appropriate data types and column naming
- → Create a separate date dimension table (date, week, month, quarter, year)
- → Used a simple star schema data model with sales as the fact table and campaign as the dimension table.
- → Write a query for Power BI: aggregated performance by platform, channel, region, and month
- → Write a query for the AI tool: flexible enough to filter by date range, platform, region, or campaign
- → Add indexes on columns you expect to filter frequently — explain why
- → Include all SQL in a single schema.sql file with comments.

★ We read SQL like we read code. Naming, structure, and comments matter as much as whether the queries return correct results.

TASK 4

Power BI — dashboard & DAX

Power BI Desktop

Connect Power BI directly to your SQL database and build a 3-page marketing performance report. No CSV imports — the SQL connection is mandatory.

- → Connect to the SQL database as the data source (not a CSV file)
- → Use the date dimension table to build a proper date relationship in the data model
- → Write DAX measures for: Total Spend, Total Sales, ROI Total Conversions, CTR %, CPC, ROAS, and, Countrywise Performance, MoM spend change
- → Page 1 — Executive summary: KPI cards + spend vs conversions trend + top 5 campaigns table
- → Page 2 — Channel breakdown: performance by platform (bar), channel mix (donut), region matrix
- → Page 3 — Audience insights: conversion rate by segment, scatter of spend vs conversions
- → At least one cross-page slicer (date + Brand Name) and one drill-through
- → Submit the .pbix file and a PDF export of all 3 pages

★ DAX quality and report layout are weighted heavily. Well-named measures, a clean field list, and readable visuals matter as much as correct numbers.

TASK 5

AI insight tool — LLM over SQL

Python · LLM API

Build a Python tool that queries the SQL database dynamically and uses an LLM to answer natural language questions about campaign performance and Sales Performance

. The AI reads from the same SQL source as Power BI.

- → Use any LLM API (OpenAI, Anthropic, Gemini) — free tier is fine
- → Accept a question via CLI or simple UI (Streamlit / Gradio is fine)
- → Translate the question into a targeted SQL query — don't dump the whole table into the prompt
- → Inject the query result as context into the LLM prompt, then return a plain-English answer
- → Must handle: "Which campaign had the worst CPC in March?" and "Summarise UK region performance"
- → Include a README.md with setup steps, design decisions, and 10 example questions
- → Within-task bonus: add conversation memory so follow-up questions work

→ The best implementations use the question to generate a SQL query first, run it, then pass only the result rows to the LLM — not the full dataset.

📌 Assignment Requirement: Video Explanation (Mandatory)

🎥 Video Submission (Compulsory)

Along with your assignment, you are required to submit a **video explanation (5–10 minutes)** explaining your complete solution.

BONUS — OPTIONAL

Go further (pick one)

Optional

One well-executed bonus beats three half-finished ones.

- → Text-to-SQL: let the AI tool auto-generate the SQL query from the user's question using the LLM, rather than hand-written query templates
- → Embed the AI tool inside Power BI using the Python visual — one interface for both
- → Anomaly detection: flag campaigns with statistically unusual spend or CPC and surface them in Power BI as a dedicated alert page
- → Budget optimiser: given a total budget, suggest channel allocation based on historical ROAS from the SQL data

SCORING RUBRIC

Category	Points	What we look at
Python — data cleaning	**20 pts**	Issue detection completeness, fix strategy, output quality, quality report clarity
Airtable Task	**10 Pts**	This task is mandatory.
SQL — schema & queries	**20 pts**	Schema design, query correctness, index choices, code readability and comments
Power BI — dashboard & DAX	**20 pts**	SQL connection used, data model quality, DAX measures, report layout and storytelling
AI tool — code & prompt design	**30 pts**	Works end to end? SQL-to-context approach, prompt quality, code structure
Pipeline integrity & communication	**15 pts**	Full pipeline cohesion, clear README, consistent data from SQL to both outputs
Bonus (if attempted)	**+20 pts**	Text-to-SQL and Power BI + AI integration score highest if working end to end

SUBMISSION INSTRUCTIONS

→ Submit as a GitHub repo (preferred) or a ZIP folder
→ Folder structure: /data, /python, /sql, /powerbi (pbix + PDF export), /ai_tool, README.md
→ Include requirements.txt — we will run your Python code
→ Power BI Desktop is free at powerbi.microsoft.com — SQLite works for the DB if no Postgres setup
→ Deadline: 3 - 4 days late submissions accepted but noted

ℹ We evaluate thinking as much as output. Comment your SQL, explain your DAX measure logic, and note why you made key prompt engineering decisions. A candidate who explains their reasoning scores higher than one who just ships code.